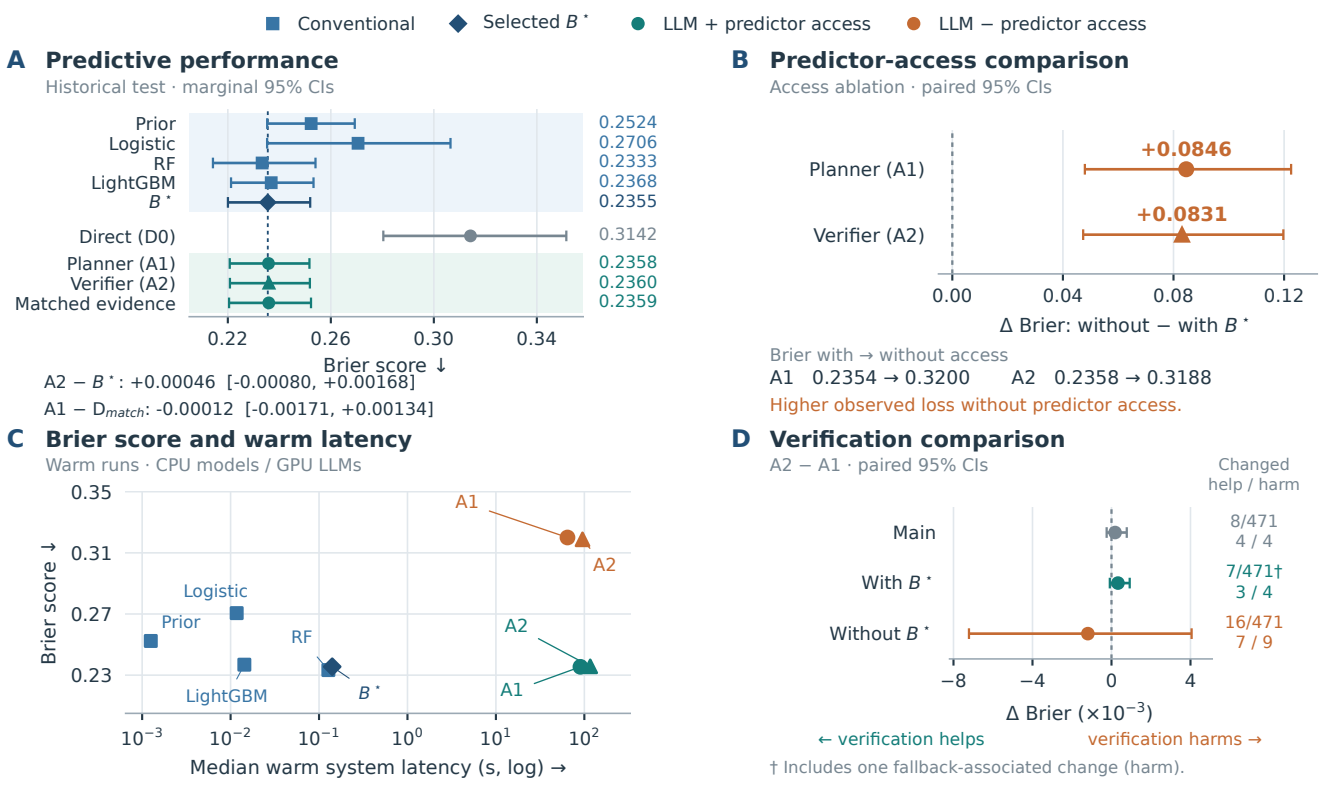
Conventional
Selected B*
LLM + predictor access
LLM − predictor access
A Predictive performance
Historical test · marginal 95% CIs
Prior
Logistic
RF
LightGBM
B*
Direct (D0)
Planner (A1)
Verifier (A2)
Matched evidence
0.2524
0.2706
0.2333
0.2368
0.2355
0.3142
0.2358
0.2360
0.2359
0.22
0.26
0.30
0.34
Brier score ↓
A2 − B*: +0.00046 [-0.00080, +0.00168]
A1 − D_match: -0.00012 [-0.00171, +0.00134]
B Predictor-access comparison
Access ablation · paired 95% CIs
Planner (A1)
Verifier (A2)
+0.0846
+0.0831
0.00
0.04
0.08
0.12
Δ Brier: without − with B*
Brier with → without access
A1 0.2354 → 0.3200
A2 0.2358 → 0.3188
Higher observed loss without predictor access.
C Brier score and warm latency
Warm runs · CPU models / GPU LLMs
Brier score ↓
0.35
0.31
0.27
0.23
Logistic
Prior
RF
LightGBM
B*
A1
A2
A2
A1
10^−3
10^−2
10^−1
10^0
10^1
10^2
Median warm system latency (s, log) →
D Verification comparison
A2 − A1 · paired 95% CIs
Changed help / harm
Main
With B*
Without B*
8/471 4 / 4
7/471† 3 / 4
16/471 7 / 9
−8
−4
0
4
Δ Brier (×10^−3)
← verification helps
verification harms →
† Includes one fallback-associated change (harm).